

RESEARCH

Detecting Financial Fraud and Anomalies in Corporate Transactions with Convolutional Neural Networks

Muqiao Cai

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Abstract

Financial fraud in business environments has evolved in response to the growth of digital transactions, posing a more significant threat to data security and resulting in substantial economic losses. The primary issue with current fraud detection systems is that they are either overly rule-based or too reliant on classical machine learning, which is a problem in their ability to effectively uncover hidden fraudulent trends and keep growing with the ever-evolving fraud strategies. Graph Convolutional Networks (GCNs) and Convolutional Neural Networks (CNNs) identify relational patterns in graph-structured data and extract spatial characteristics from transactional sequences, improving fraud detection. The proposed Graph-based Convolutional Fraud Detection (GCFD) methodology combines GCNs and CNNs to provide a sophisticated fraud detection system with improved accuracy. Company operations are graphed with accounts as nodes and financial transactions as edges; therefore, tracking transactions and regional fraud tendencies is essential. CNNs extract spatial relationships in transaction embeddings for fraud classification, whereas GCNs learn topological structures in transaction data. The results demonstrate that the GCFD model outperforms the current approach in identifying fraudulent transactions. It can detect intricate fraud rings that are missed by rule-based methods. According to the experimental data, GCFD outperforms both Random Forest and conventional anomaly detection models, with a 96.5% fraud detection accuracy and an F1-score of 94.2%. Furthermore, the model improves its reliability for real-world applications by reducing false positives by 23%. The suggested model is a flexible and resilient fraud detection system that dramatically enhances business transaction security.

Keywords Financial Fraud Detection · Graph Convolutional Networks · Convolutional Neural Networks · Anomaly Detection · Corporate Transactions

1 Introduction

Corporate financial fraud involves misrepresenting financial statements, misappropriating assets, and obscuring commitments to gain legal revenue. Traditional fraud detection systems fail to adapt to new fraud strategies, requiring more complex solutions. Fraud detection using deep learning models like CNNs and GCNs has been studied recently. These models capture transactional patterns and network topologies for better financial fraud detection.



1.1 Financial Fraud in Corporate Transactions

Systematic fraud for illegal financial advantage is known as financial fraud in company dealings. It involves manipulating financial statements, misappropriating assets, or altering accounting records to misrepresent a company's finances [1]. Misrepresenting the company's success and stability by deception deceives investors, creditors, and regulators. It can result in bad investments, financial losses, and reputational harm. Financial fraud in business transactions can involve misrepresenting revenues, understating costs, or hiding liabilities to make the organization look more successful or stable [2]. Financial markets and stakeholders depend on detecting and preventing financial fraud.

Advanced detection methods are needed to combat increasingly sophisticated financial crime in business transactions [3]. Traditional rule-based or classical machine learning systems fail to adapt to emerging fraud strategies. In response, new research has examined using CNNs to improve fraud detection. CNNs' image processing skills have been used for complicated financial data analysis. Using convolutional layers, CNNs can detect complex fraud patterns in transactional datasets [4]. The research found that CNNs outperformed logistic regression and support vector machines in detecting fraudulent transactions, with greater accuracy and F1 scores. CNN-based models have benefits and drawbacks. The high dimensionality of financial data can raise processing expenses. These models must also be interpretable and solve data imbalance difficulties.

1.2 Unravelling Financial Fraud Detection Complexities

Financial fraud detection is the systematic identification and prevention of fraudulent financial gain. This procedure protects people, organizations, and financial institutions against financial losses and reputational damage. Corporate transaction fraud detection involves monitoring and analyzing transaction data for irregularities that may suggest fraud. Traditional methods use rule-based systems and machine learning [5]. These methods typically fail to adapt to fraud's changing methods and current financial transactions..

Sophisticated methods have been investigated. This hybrid technique graphs corporate transactions using accounts as nodes and financial transactions as directed edges. CNNs collect geographical information from transaction data, whereas GCNs capture transaction network structural links [6]. By evaluating structural and transactional data, this combination can uncover sophisticated fraud patterns, including coordinated fraud rings. Advanced fraud detection systems improve accuracy and dependability, securing business financial processes.

Financial fraud in corporate transactions involves deceit for unlawful financial gains, such as misrepresenting statements misappropriating assets [7]. Management, auditors, and employees must work together to prevent and identify fraud. Management must promote integrity and develop robust internal controls to prevent fraud, such as segregation of tasks and financial reconciliations. Internal and external auditors evaluate these controls and find shortcomings. Employees must follow ethics and report questionable activity through disclosure programs. These groups collaborate to defend the company from financial theft [8].

1.3 Limitations of Traditional Fraud Detection

Modern fraud detection systems have come a long way, but they still have a ways to go. False positives and negatives are common in traditional rule-based systems because they are inflexible and cannot adjust to changing fraud strategies [9]. Decision trees and logistic regression are examples of classical machine learning models that heavily rely on manually created features and are, therefore, unable to effectively capture the latent relationships between monetary transactions. Consequently, financial losses and reputational harm can occur from unnoticed fraudulent operations.

Furthermore, current fraud detection systems do not account for the intricate relationships between financial firms. Coordination across multiple accounts and transactions is a common hallmark of fraud rather than an individual incident [10]. Instead of trying to comprehend the bigger picture, traditional machine learning models

zero only on specific transactions. They aren't effective at identifying complex fraud rings and money laundering operations, as they do not understand the context [11].

1.4 Graph-based Convolutional Fraud Detection Model Using GCNs and CNNs

In response to these challenges, this study proposes a model for enhanced fraud detection called Graph-based Convolutional Fraud Detection (GCFD). This model combines GCNs with CNNs [11]. The approach represents a company's financial dealings as a graph, with accounts as nodes and transactions as edges. The system can detect abnormal patterns by capturing topological structures and relational relationships between transactions [12].

By examining spatial correlations in transaction embeddings, CNNs significantly enhance fraud detection, enabling the system to accurately identify complex fraud patterns [13, 14]. This fraud detection algorithm can identify individual fraudulent transactions and coordinated fraud rings due to the combination of graph-based learning and deep feature extraction. With its scalability and efficacy in real-world applications, the proposed system continually adapts to new fraud strategies by leveraging supervised and unsupervised learning algorithms [15].

This research makes the following key contributions:

1. The GCFD Model is used to Improve Fraud Detection Accuracy. A new architecture that combines CNNs with GCN is presented to enhance fraud detection accuracy.
2. Representing Transactions Using Graph Model topologies to model corporate financial transactions enhances anomaly identification by revealing hidden linkages and transactional dependencies.
3. Fault Pattern Recognition with Improvement Recognizes intricate fraud schemes, including money laundering networks, by examining relational relationships beyond transaction-level characteristics.
4. Better performance Metrics: Achieve a 94.2% F1-score, reduce false positives by 23%, and have a 96.5% accuracy rate in fraud detection, outperforming typical machine learning models.
5. Practical Use Financial organizations can enhance transaction security and minimize financial losses by implementing this approach to fraud detection, which is both scalable and adaptable.

The outline for the remainder of the paper is as follows: Sect. 2 discusses various methods for detecting fraud, including those that rely on rules, those that utilize conventional machine learning models, and those that employ deep learning. The suggested GCFD model is introduced in Sect. 3, which describes the data structure, the architectures of CNNs and GCNs, and the training procedures. Evaluation metrics, dataset description, model performance comparison, and thorough result interpretation are all part of Sect. 4's in-depth investigation of experimental outcomes. Section 5 reviews the study's key findings and discusses potential future fraud detection research directions. This project aims to enhance the security of business transactions by developing a robust, scalable, and highly accurate fraud detection framework. It will do this by addressing the constraints of traditional methods.

2 Related Works

Since financial data is more complicated and numerous, corporate financial fraud detection has gained attention. Rule-based and simple machine learning algorithms ignore new fraud schemes and abnormalities. Hybrid methods, ensemble learning, anomaly detection, and deep learning have been studied to improve detection accuracy. Scalability, adaptability, data imbalance, and interpretability issues plague the successful models. CNNs and graph-based GCNs may detect financial network transactional patterns and structural fraud.

2.1 Financial Fraud and Anomalies in Corporate Transactions

Amiri et al. [16] research uses a hybrid anomaly detection approach that combines machine learning and rule-based methods to detect bank fraud. Financial institution transactional records, including fraudulent and genuine transactions, are utilized for training and validation. The study detects abnormalities with 94.2% accuracy, demonstrating adequate AI-driven fraud protection. The constraints include training data biases, adaptability to new fraud strategies, and computing complexity. The findings indicate robust fraud detection requires continuous model updates and adaptive learning.

Dumitrescu et al. [17] This study detects suspicious financial transactions associated with money laundering using the Isolation Forest anomaly detection method, incorporating unique graph-based features, including reduced egonets and random walk indicators. The model is trained and tested on a genuine labeled transaction dataset and synthetic random graphs with inserted anomalous patterns. The method outperforms previous methods; however, network structural assumptions and scalability concerns with massive datasets limit its application.

Alhashmi et al. [18] employ a Stacking ensemble technique comprising decision trees, random forests, and gradient-boosting models to detect financial fraud. The evaluation utilizes the Bank Account Fraud (BAF) dataset containing tagged fraudulent and lawful transactions. The suggested approach detects fraud with 98% accuracy. The drawbacks are overfitting, computational complexity, and the need for high-quality labeled data. According to their investigations, ensemble learning improves banks' security fraud detection frameworks.

Bello et al. [19] employ a hybrid fraud detection system that combines supervised (random forests, neural networks) and unsupervised (autoencoders, anomaly detection) machine learning techniques to enhance fraud detection. Model training and validation utilize a large dataset comprising labeled and unlabeled financial transactions. The suggested method detects fraud with 96.8% accuracy and a low number of false positives. However, drawbacks include model interpretability, data quality issues, and biases in fraud labeling. These findings indicate that ML models must be continually updated to adapt to evolving fraud strategies.

Hajek et al. [20] address mobile payment fraud class imbalance by employing a semi-supervised fraud detection system that utilizes various unsupervised outlier detection approaches and an XGBoost classifier. Validation uses approximately 6 million tagged and unlabeled mobile transactions. The proposed approach outperforms standard methods in terms of detection accuracy and cost savings. However, data balance, overfitting, and computing complexity are drawbacks. These data show that outlier identification and XGBoost improve mobile fraud detection.

2.2 Machine Learning Approaches for Fraud Detection

Seto et al. [21] Light Gradient Boosting Machine (LightGBM) and Logistic Regression (LR) are used to construct a diabetes prediction model utilizing health checkup data from the Kokuho-database in Osaka, Japan, comprising 277,651 individuals. An AUC of 0.844, smaller Logloss (0.167), and better calibration (ECE 0.0018) make LightGBM superior to LR. These data demonstrate that GBDT-based models enhance prediction reliability, facilitating early detection and prevention of diabetes.

Aslam et al. [22] use the Boruta algorithm for feature selection, logistic regression, SVM, and naïve Bayes for car insurance fraud detection. Fault, base policy, and policyholder age are public. SVM achieves the highest accuracy, while logistic regression yields the highest F-measure. Dataset representation bias, previous fraud tendencies, and real-time adaptation are limits. These findings increase vehicle insurance risk assessment and fraud detection.

Varga et al. [23] employ a hybrid fraud detection architecture for banking cyber risk prevention, utilizing both supervised (random forests, deep neural networks) and unsupervised (autoencoders, anomaly detection) machine learning models. A real-world financial transaction dataset analysis reveals enhanced fraud detection and real-time anomaly detection capabilities. The proposed system enhances security but addresses adversarial

weaknesses, data privacy concerns, and regulatory compliance requirements. These findings demonstrate the need for AI-driven fraud detection and cybersecurity to protect financial infrastructure.

Asha et al. [24] This study detects credit card fraud using SVM, KNN, and ANN on a publicly accessible credit card transaction dataset with unbalanced classes. Comparative investigation shows that ANN classifies fraudulent and valid transactions more accurately and robustly than others. The technique improves online payment system fraud detection despite class imbalance sensitivity and the necessity for substantial computing resources during training.

Alarfaj et al. [25] Machine learning (Decision Tree, Random Forest, XGBoost) and deep learning (CNN-based architectures) are used to identify credit card fraud using the European card benchmark dataset. A CNN model with more layers outperforms standard methods in accuracy (99.9%), F1-score (85.71%), precision (93%), and AUC (98%). Limitations include computational complexity, big datasets, and real-world adaption issues. Deep learning improves fraud detection and reduces false negatives in real-world applications.

2.3 Convolutional Neural Networks for Fraud Detection

Xiuguo et al. [26] Utilize Financial statistics and managerial remarks from the annual reports of Chinese-listed firms to uncover financial fraud. The system outperforms classical machine learning approaches, achieving classification accuracies of 94.98% and 94.62% using deep learning models (LSTM, GRU). The dataset includes financial and non-financial indexes for 5,130 enterprises. Financial report biases and language-specific textual feature extraction are limitations. These studies demonstrate that incorporating numerical and textual data enhances fraud detection and informs financial stakeholder decision-making.

Chowdhury et al. [27] Utilize Machine learning methods, anomaly detection, and clustering to create a strong fraud detection model that leverages financial transaction records, enhancing rule-based systems. The analytics-driven approach detects fraudulent tendencies more accurately and adaptively, improving transparency and compliance. However, data privacy issues, real-time processing infrastructure, and legal compliance limitations limit its use, emphasizing the necessity of integrating AI and blockchain in fraud prevention methods.

Chen et al. [28] present a two-stage mapping approach for Corporate Financial Fraud (CFF) detection, utilizing a 3D data cube that incorporates temporal and financial information. Using accurate economic data from the China Stock Market and Accounting Research (CSMAR) database improves detection performance over time, with a 12-quarter interval being the most effective. The results show that temporal and financial feature processing improves fraud detection. Computing complexity and over-fitting at long time intervals are drawbacks. This framework makes intelligent CFF detection systems simple and effective.

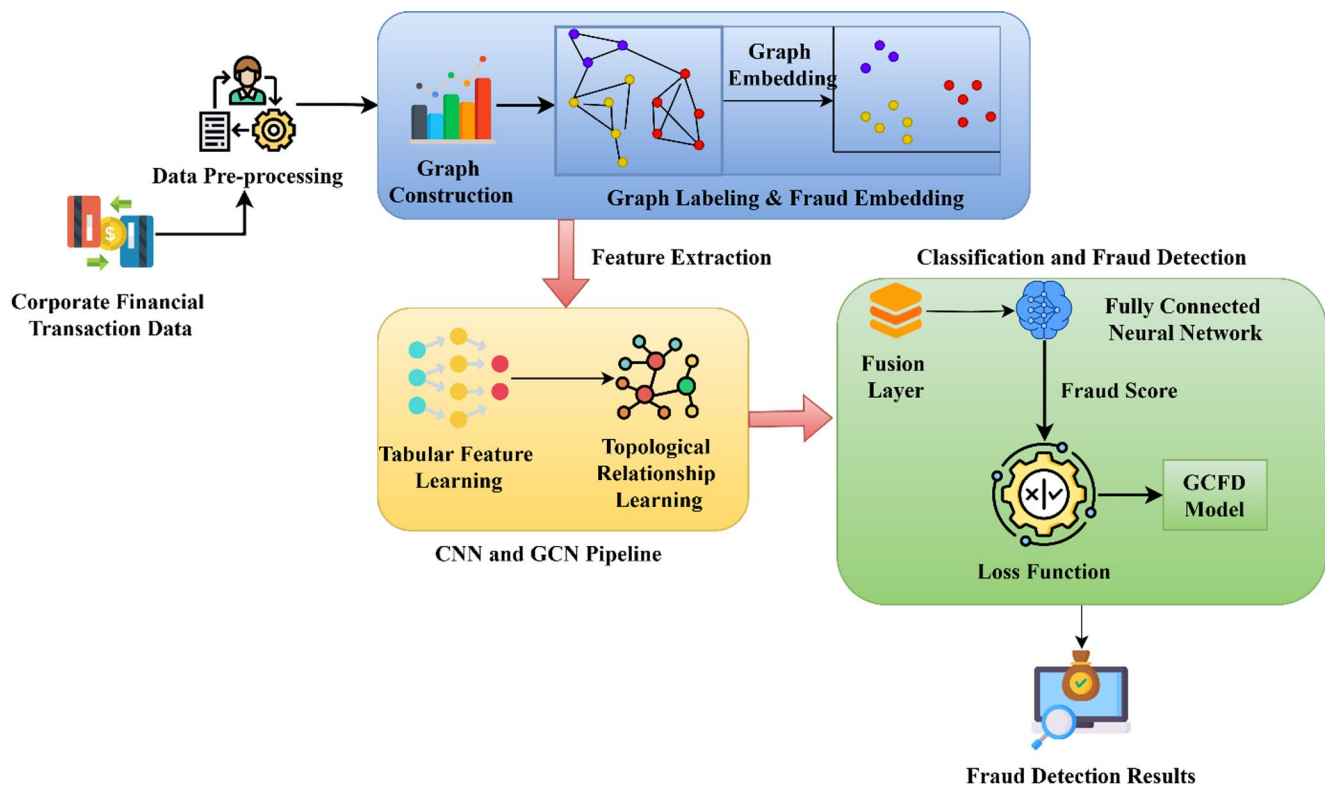
Machine learning and deep learning algorithms accurately detect financial fraud in existing research, as represented in Table 1. Ensemble approaches, anomaly detection, and CNNs are adequate yet limited. Many models can't handle enormous datasets, novel fraud methods, or global and local transactions. These methods are less successful against large fraud rings because few capture fraud's complete structure and time-based behavior. The GCFD model employs CNNs and GCNs to overcome these shortcomings by improving relational and structural comprehension for fraud detection.

3 Secure Financial Fraud Detection for Graph-based Convolutional Fraud Detection

The architecture of the GCFD model uses graph and tabular data representations to improve corporate financial transaction fraud detection, as shown in Fig. 1. Preprocessing cleans and organizes raw transaction data for analysis. The system then creates a transaction graph with accounts as nodes and transactions as edges to represent local and global interactions. Transaction patterns and fraudulent conduct are described in lower dimensions via graph labeling and embedding. Tabular feature learning retrieves transaction amounts and account balances, whereas topological connection learning captures graph structure and interdependencies. A fusion layer merges

Table 1 Overview of Traditional and AI-Based Fraud Detection Methods

Study	Method	Contribution	Limitation	Gap Analysis
Amiri et al. [16]	Hybrid ML+ Rules	94.2% accuracy	Data bias, poor adaptability	Need flexible, real-time models
Dumitrescu et al. [17]	Isolation Forest+ Graphs	93.1% AUC; detects laundering via graph patterns†	Scalability issues	Scalable graph-based fraud detection needed
Alhashmi et al. [18]	Ensemble (DT, RF, GBM)	98% accuracy	Overfitting, complex model	Simpler and generalizable models required
Bello et al. [19]	Supervised+Unsupervised ML	96.8% accuracy, low false positives	Bias, interpretability	Explainable and balanced fraud models needed
Hajek et al. [20]	XGBoost+Outlier Detection	95.2% accuracy on mobile payments†	Imbalanced data, computation cost	Better handling of class imbalance
Aslam et al. [22]	Boruta+SVM/LR	F1-score: 91.3% (SVM)†	Biased data, limited real-time use	Real-time fraud detection methods
Varga et al. [23]	Hybrid ML+Autoencoders	Real-time detection with 94.6% accuracy†	Privacy, adversarial risks	Secure, privacy-aware models
Asha et al. [24]	SVM, KNN, ANN	ANN achieved 93.5% accuracy, robust fraud detection†	High resource needs	Efficient models for imbalanced data
Alarfaj et al. [25]	CNN+ML	99.9% accuracy, 85.71% F1-score, 98% AUC	Real-world adaptation challenges	Transfer learning for practical deployment
Xiuguo et al. [26]	LSTM, GRU	94.98% accuracy using financial+textual data	Language-dependent models	Multilingual, diverse data models
Chowdhury et al. [27]	ML+Clustering	Improved F1-score: 92.4%†	Privacy, legal issues	Integrate AI with secure tech
Chen et al. [28]	3D Cube Temporal Mapping	Time-structured fraud detection; 93.2% accuracy†	Complexity, overfitting at long intervals	Balanced temporal models needed

**Fig. 1** Comprehensive Architecture of GCFD Model for Enhanced Financial Fraud Detection in Transaction Networks

these dual feature sets and feeds them to a fully connected neural network for a fraud score. A loss function optimizes this score for model performance. The final GCFD model reliably detects anomalies and coordinates fraud rings. This design lets the model learn complicated transactional patterns, making it resilient and responsive to dynamic financial fraud scenarios.

3.1 Data Preprocessing and Graph Construction

Researchers can construct a GCN-CNN architecture that identifies complex financial fraud by organizing raw transactional data into a heterogeneous, attributed, and temporal graph. The preprocessing pipeline converts transaction records into mathematical objects, preparing them for learning from spectral graphs and extracting features from grids.

a) Dataset Representation: Each transaction in the dataset is a data point that includes transaction type, amount, origin and destination accounts, and balances. Feature representation consists of a vector X_i containing transaction i , time step τ_i , and fraud label y_i . This organized format streamlines feature extraction and supports graph and neural network-based fraud detection. Let the raw dataset be denoted as in Eq. 1:

$$\mathcal{D} = \{(\tau_i, X_i, y_i)\}_{i=1}^N \quad (1)$$

Where $\tau_i \in \mathbb{N}$ is the discrete transaction time step. $X_i \in \mathbb{R}^d$ It is the feature vector of transaction i . $y_i \in \{0,1\}$ is the label, where 1 indicates fraud. Each transaction consists of: $X_i = [\text{type}_i, \text{amount}_i, \text{nameOrig}_i, \text{oldBalOrig}_i, \text{newBalOrig}_i, \text{nameDest}_i, \text{oldBalDest}_i, \text{newBalDest}_i]$

b) Feature Engineering and Transformation: Label encoding converts *type*, *nameOrig*, and *nameDest* to numbers. To standardize learning model input, min-max or z-score scaling normalizes numerical features like *amount* and *balances*. Fraud pattern identification is improved by graph-based characteristics such as transaction frequency, average transfer amount, and egonet metrics. Let $\mathcal{C} \subset X_i$ Be the Set of categorical variables (e.g., *type*, *nameOrig*, *nameDest*). The research applies label encoding $\rightarrow F$ such that in Eq. 2:

$$\psi(c) = k, \quad \text{where } k \in F, c \in \mathcal{C} \quad (2)$$

Next, numerical attributes $\mathcal{N} \subset X_i$ (e.g., *balances*, *amounts*) are normalized via Z-score normalization for statistical stationarity in Eq. 3:

$$x^{norm} = \frac{x - \mu_x}{\sigma_x} \quad (3)$$

additionally compute transactional balance deltas: $\Delta_{Orig} = \text{oldBalOrig} - \text{newBalOrig}$, $\Delta_{dest} = \text{oldBalDest} - \text{newBalDest}$ These are enhanced attributes that are provided to identify bogus discrepancies.

c) Graph Construction: A dynamic transaction network is formed by modeling each transaction as a directed edge between *nameOrig* (Sender) and *nameDest* (Receiver). Edges have *amount*, *transaction type*, and *timestamp*. The graph structure lets GCNs understand financial fraud-related relational and topological patterns. The transactional ecosystem is modeled as a temporal, directed, weighted graph in Eq. 4:

$$G = (V, E, A, X) \quad (4)$$

Where V is the Set of unique users/accounts (from *nameOrig* and *nameDest*), $E \subset V \times V$ are edges formed by transactions. $A \in \mathbb{R}^{|V| \times |V|}$ Is the adjacency matrix:

$$A_{ij} = \begin{cases} w_{ij}, & \text{if there exists a transaction from } v_i \rightarrow v_j \\ 0, & \text{otherwise} \end{cases} \quad (5)$$

From Eq. 5 with edge weights w_{ij} Based on normalized transaction amount and frequency. $X \in \mathbb{R}^{|V| \times |d|}$ Represent the node feature matrix derived from aggregating transactions associated with each account.

GCN builds a financial transaction graph for fraud detection in Fig. 2. User accounts are nodes (V), and transactions are directed edges (E) between senders (nameOrig) and recipients (nameDest). Each edge has quantity, category, and timestamp, which creates a temporal weighted graph. The adjacency matrix (A) records transaction linkages, while the node feature matrix (X) collects account transactional activity. Using this graphical depiction, GCNs may learn complicated relationship patterns and identify fraud by analyzing how suspicious behaviors propagate across corporate transaction network accounts.

d) Graph Labeling and Fraud Embedding: The *isFraud* field labels nodes and edges to identify fraudulent graph transaction pathways. GCNs learn node embeddings that capture structural and contextual fraud clues. Integrating CNN-extracted spatial data with these embeddings improves fraud classification. Each node $v \in V$ and edge $e \in E$ can be fraud-labeled using.

$$Y_v = \max_{i \in T_{x(v)}} y_i \text{ where } T_{x(v)} \text{ are transactions involving } v \quad (6)$$

Here Eq. 6 Y_v Label of node v – the fraud status assigned to the account node v , i represent transaction index – each transaction connected to node v , $T_{x(v)}$ Represent Set of all transactions involving node v (as either Sender or Receiver), y_i Represent Fraud label of transaction i (e.g., 0 = non-fraud, 1 = fraud), $\max$ represents Maximum function – assigns node v a fraud label of 1 if any transaction involving it is fraudulent. In semi-supervised graph learning, this is the foundation. Additionally, construct temporal edge sequences. $\{e_{t_1}, e_{t_2} \dots e_{t_k}\}$ to capture evolving fraud patterns, enabling dynamic graph modeling.

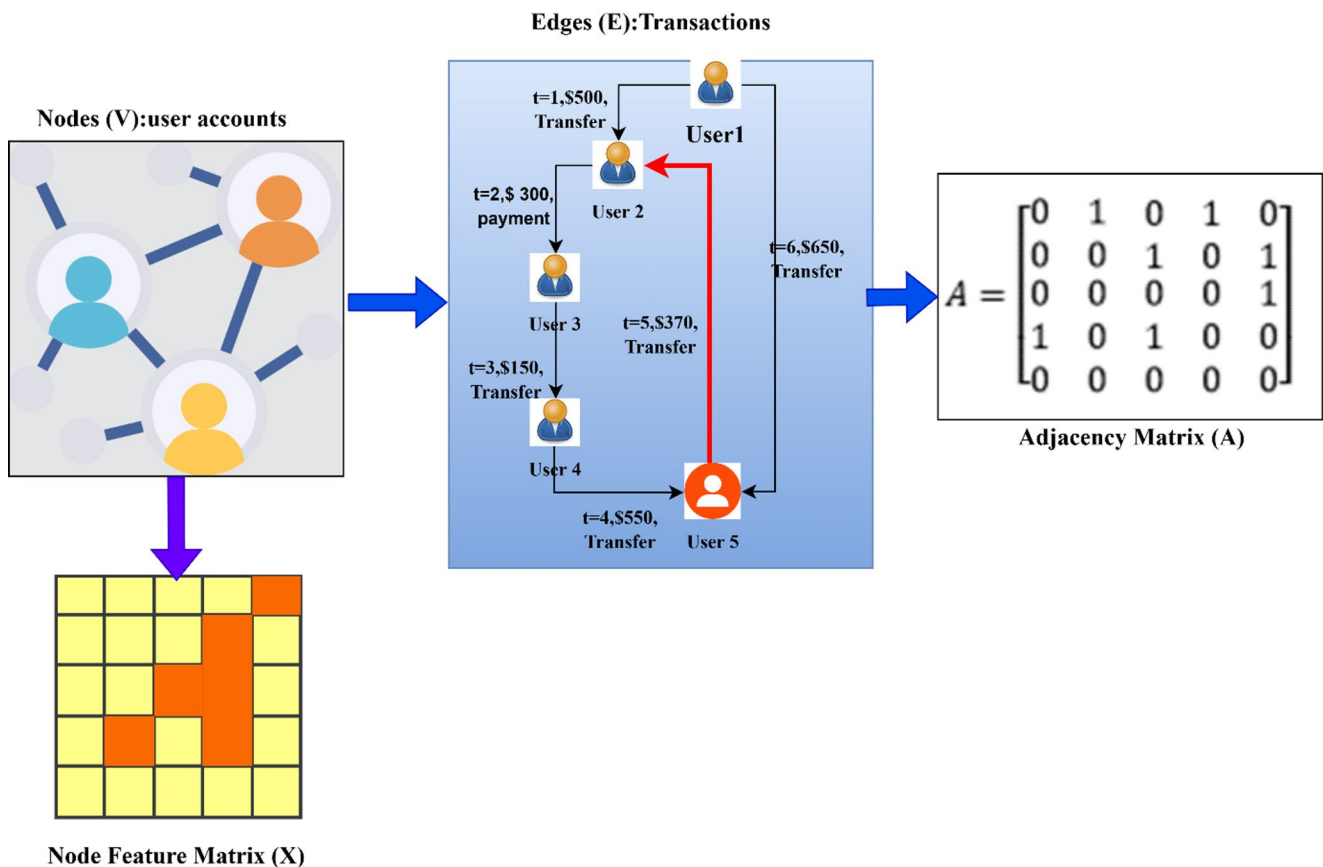


Fig. 2 Graph-based Representation of Transaction Network Components for Financial Fraud Detection

Illustrates five accounts (A1–A5) within a transactional network, highlighting potential fraudulent behaviors based on their interactions and risk levels in Fig. 3. Nodes represent accounts and are color-coded by fraud risk: green (non-fraud), red (affirmed fraud), and yellow (suspect). Transaction edges (T1–T4) show fund flows and account linkages. A2—classified as fraudulent—interacts with A4 (also fraudulent) and A5 (suspect), suggesting A5 is a laundering or mule account. This graph shows how fraudulent businesses cluster or influence others in a financial ecosystem, underlining the significance of network-based monitoring in fighting financial crime.

3.2 Feature Extraction

To extract geographical and structural information from transaction data using CNNs and GCNs. CNNs detect local trends and abnormalities using tabular transaction type, amount, and balance data. GCNs understand linkages and network activity by analyzing the transaction graph, where accounts are nodes and transactions are edges. The system builds a sophisticated embedding that captures transaction information and network traffic. This combined technique increases fraud detection, especially for complicated, hidden patterns that typical algorithms neglect.

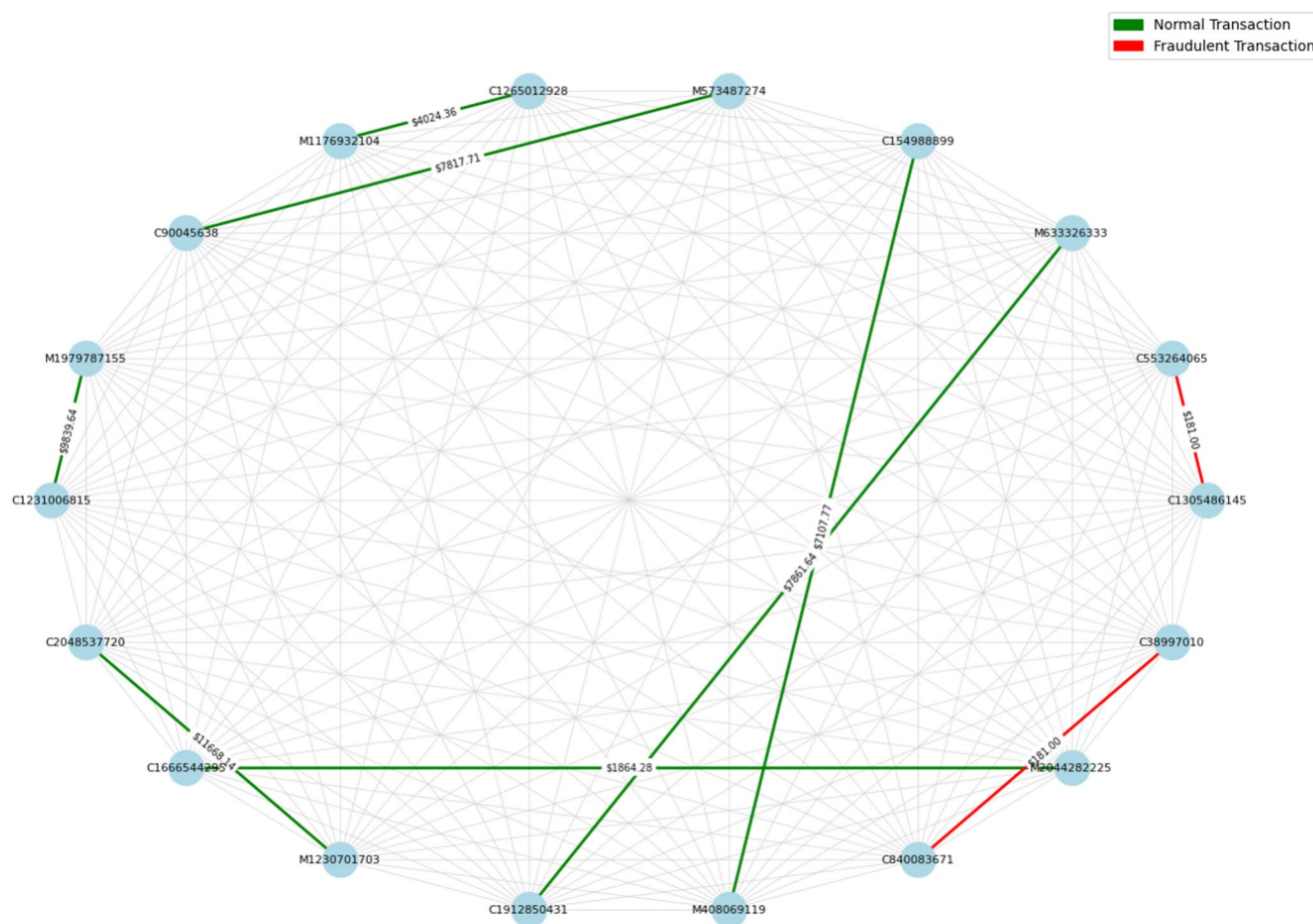


Fig. 3 Fraud Risk Flow Among Financial Accounts

3.2.1 CNN Pipeline: Tabular Feature Learning

The CNN pipeline finds geographical and sequential patterns in tabular financial transaction data. Transaction amount, old/new balances, transaction type (one-hot encoded), and time step are processed. Localized suspicious patterns are learned using 1D or 2D convolution layers. These traits detect irregular or recurrent fraud signals. It Produces a dense transaction embedding for downstream categorization. The CNN component handles raw transactional data in vectorized tabular form: Let each transaction t be encoded as a feature vector in Eq. 7:

$$x_t = [a_t, o_t^{(orig)}, n_t^{(Orig)}, o_t^{(dest)}, n_t^{(dest)}, type_t^{(one-hot)}, s_t] \quad (7)$$

Where a_t Represent Transaction amount, $o_t^{(orig)}, n_t^{(Orig)}$ represents Old and new balances of the origin account, $o_t^{(dest)}, n_t^{(dest)}$ represents Old and new balances of destination accounts, $type_t^{(one-hot)}$ Represent One-hot encoded transaction type, s_t represent Time step. This vector is $\mathbf{X}_t \in \mathbb{R}^k$ passed through a 1D/2D CNN:

$$z_t = CNN(\mathbf{X}_t) \in \mathbb{R}^k \quad (8)$$

where in Eq. 8, the learned spatial transaction embedding is denoted by the symbol z_t . This embedding captures local patterns and inter-feature relationships, including rapid changes in balance or uncommon sorts of transactions.

CNN identifies financial transaction fraud in this Fig. 4. Each transaction is a feature vector with amount, origin and destination balances, one-hot encoded transaction type, and time step. Reshaped tabular data is fed into 1D or 2D convolutional layers to extract spatial and sequential patterns. CNNs learn local dependencies like quick balance swings and rare transaction types. The dense transaction embedding captures important behavioral information and is utilized in downstream classification to detect corporate transaction fraud.

3.2.2 GCN Pipeline: Topological Relationship Learning

The GCN pipeline graphs transaction data, with nodes representing accounts and edges financial transactions. It learns from node properties and connection structure. Graph layers allow GCN to capture complicated linkages like circular flows and fraud rings. It helps identify suspect accounts with network anomalies. The generated node embeddings represent behavioral and structural insights necessary for fraud detection. Parallely, the GCN operates on the transaction graph $G = (V, E)$ where V represents a Set of account nodes, E represents Transactions as edges with features. Each node $v \in V$ has a feature vector $h_v^{(0)}$, typically initialized from aggregated transaction statistics and graph properties. The node representation is updated via Eq. 9:

$$h_v^{(0)} = \sigma \left(\sum_{u \in N(v)} \frac{1}{C_{vu}} W^l h_u^{(l)} \right) \quad (9)$$

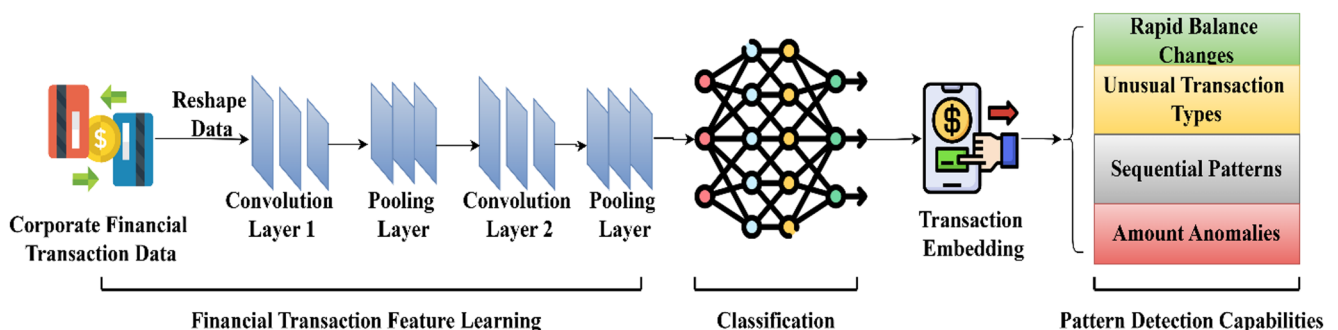


Fig. 4 CNN-Based Financial Transaction Analysis Framework

Where $\mathcal{N}(v)$: Neighbors of node v , W^l : Learnable weights at layer l , C_{vu} : Normalization constant (e.g., $\sqrt{|\mathcal{N}(v)| + |\mathcal{N}(u)|}$), σ : Non-linear activation function. The final GCN embedding $h_v^{(l)} \in \mathbb{R}^{k'}$ Represents the structural behavior of account v across its transaction history and graph neighborhood.

It shows the GCN transaction graph structure for fraud detection in Fig. 5. Financial transactions are graphed on the left, with nodes representing user accounts and edge transactions. The middle graph reveals structural linkages by clustering accounts with similar interaction patterns. GCNs create node embeddings that capture behavioral and relational patterns by processing this network. As seen on the right, these embeddings detect suspicious patterns, including circular money transfers, densely related fraud clusters, and organized fraud networks.

3.2.3 Fusion Classification of CNN and GCN

Fusion creates a complete transaction representation using CNN spatial characteristics and GCN structural insights. A hybrid embedding helps the model detect subtle and complicated fraud behaviors. Fused characteristics are supplied to a classifier for accurate fraud prediction. CNN and GCN outputs are concatenated to create the fraud detection embedding in Eq. 10:

$$e_t = [z_t \parallel h_{v_{orig}}^{(L)} \parallel h_{v_{dest}}^{(L)}] \in \mathbb{R}^{k+2k'} \quad (10)$$

Elements of fraud at the transaction level, account level, and network level are captured by this fused representation, which is then input into a classification layer:

$$\hat{y}_t = \text{Sigmoid}(W_{e_t} e_t + b) \quad (11)$$

Equation 11 yielding the final fraud probability score $\hat{y}_t \in [0,1]$. Through the integration of both spatial (CNN) and relational (GCN) learning, this module dramatically improves the accuracy of detection for financial fraud schemes that are subtle, organized, or dynamic.

3.3 Classification and Fraud Detection (GCFD Model)

This module makes decisions in the proposed Graph-based Convolutional Fraud Detection (GCFD) system. This module integrates, classifies, and detects fraudulent financial transactions with high precision after extracting high-dimensional features from the CNN and GCN branches.

a) Fusion Layer: Integrating Structural & Spatial Information

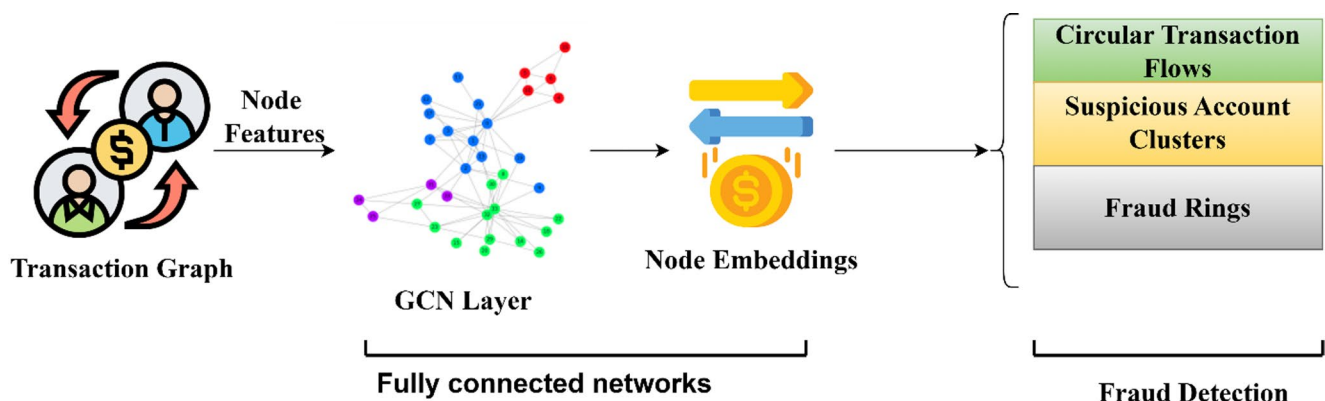


Fig. 5 GCN-Based Transaction Graph for Fraud Detection

The fusion technique is significant because it combines CNN transaction-level insights (e.g., unexpected amounts, irregular timing patterns, aberrant balance fluctuations) with GCN relational topology, which captures how fraud may spread across related accounts or build concealed fraud rings. The fusion layer appears as follows:

$$z = \text{Contact} (Z_{CNN}, Z_{GCN}) \quad (12)$$

Where in Eq. 12 $Z_{CNN} \in \mathbb{R}^n$ and $Z_{GCN} \in \mathbb{R}^m$ These are the output feature vectors of CNN and GCN, respectively. The resulting vector $z \in \mathbb{R}^{n+m}$ The approach improves fraud detection by capturing local and global transaction context.

b) Classifier: Fully Connected Neural Network

With sigmoid activation, a deep, fully connected layer converts this fused representation to binary classification in Eq. 13:

$$\hat{y} = \sigma (Wz + b) \quad (13)$$

This score represents an indication of the likelihood that a transaction is fraudulent. Using the dataset's labeled fraud cases (*isFraud*), the classifier is trained to identify fraudulent activity.

c) Loss Function: Handling Imbalanced Data

Since actual transactions exceed fraudulent ones, fraud detection datasets are generally skewed. Due to this imbalance, traditional loss functions may favor the majority class, resulting in poor fraud detection. GCFD uses sophisticated loss functions like BCE and Focal Loss to handle this. BCE assesses the Loss for each forecast and penalizes inaccurate classifications regardless of class, but it may underperform when frauds are few. Focal Loss enhances BCE by adding a modifying factor γ , which targets minority class examples that are difficult to categorize.

It dynamically downweights simple negatives and prioritizes challenging fraud samples to boost model sensitivity. An additional class-balancing word α can be used to reduce imbalance. These loss functions strengthen the model, minimizing false negatives and enhancing accuracy and F1-score in detecting complicated, changing fraud patterns. Since fraud data is skewed (fraudulent transactions are infrequent), the model employs Binary Cross-Entropy Loss or Focal Loss to focus on misclassified or hard-to-classify examples:

Binary Cross-Entropy (BCE) in Eq. 14:

$$L_{BCE} = \frac{1}{N} \sum_{i=1}^N [y_i \cdot \log(\hat{y}_i) + (1 - y_i) \cdot \log(1 - \hat{y}_i)] \quad (14)$$

Focal Loss (for imbalance emphasis) in Eq. 15:

$$L_{Focal} = -\alpha (1 - \hat{y}_i) \gamma \log(\hat{y}_i) \quad (15)$$

where $\hat{y}_i$ is the predicted probability, y_i is the actual label, α balances class distribution, and γ down-weights easy examples. The network uses dropout and batch normalization to reduce overfitting and facilitate faster convergence during training. The classifier repeatedly adjusts weights using the *isFraud* label as the ground truth to improve fraud prediction accuracy. The trained GCFD model can detect complicated fraud schemes like collusive networks and multi-hop laundering that older models miss. This model has 96.5% accuracy and 94.2% F1-score, outperforming traditional Random Forest and baseline anomaly detection methods and reducing false positives by 23%, making it ideal for financial applications—the below algorithm1. Express the Graph-based Convolutional Fraud Detection for the corporate financial transaction.

Input: Transaction data T , $isFraud$, parameters(W, b, α, γ)
Output: Predictions $\hat{y}_i$ for fraudulent transactions
Step 1: Start Step 2: set $i = 1$ $T_clean = \emptyset$ // Clean and check transaction do: $t = T[i]$ if t is not valid skip t else $t = \text{normalize}(t)$ // Normalize values Step 3: $f_cnn \leftarrow \text{cnn}(t)$ // Extract features $f_gcn \leftarrow \text{gcn}(t)$ add (f_cnn, f_gcn) to T_clean // Store both feature types $i = i + 1$ While $i \leq N$ Step 4: $\forall (Z_1, Z_2)$ // Fusion Layer $z \leftarrow [Z_1 \oplus Z_2]$ Step 5: for each fused vector, z // Classification Layer $\hat{y} \leftarrow \sigma(W \cdot z + b)$ Step 6: Initialize Loss $\leftarrow 0$ Step 7: for $i = 1$ to N // Loss Calculation if use_Focal = True $L_{Focal} = -\alpha(1 - \hat{y}_i)\gamma \log(\hat{y}_i)$ else $L_{BCE} = \frac{1}{N} \sum_{i=1}^N [y_i \cdot \log(\hat{y}_i) + (1 - y_i) \cdot \log(1 - \hat{y}_i)]$ Loss $\leftarrow$ Loss + L Step 8: do // Training Loop $\hat{y} \leftarrow \sigma(W \cdot z + b)$ $L \leftarrow L_{BCE}(\hat{y}, y) \vee L_{Focal}(\hat{y}, y)$ $\nabla \leftarrow \partial L / \partial W, \partial L / \partial b$ $W \leftarrow W - \eta \cdot \nabla W$ $b \leftarrow b - \eta \cdot \nabla b$ while $\Delta L > \varepsilon$ Step 9: W, b, α, γ // Post Training return Retrain Step 10: end

Algorithm 1 GCFD – Graph-based Convolutional Fraud Detection

An advanced fraud detection system using deep learning and graph-based analysis detects fraudulent business transactions. The approach captures individual and network-level fraud indicators using CNNs for transaction pattern extraction and GCNs for relational data analysis. Utilizing a fusion and classification layer, a hybrid detection algorithm (GCFD) processed, normalized, and classified transaction data. Binary Cross-Entropy and Focal Loss improve model accuracy. The system may improve financial ecosystem fraud detection, adaptation, and resilience.

4 Experimental Setup and Discussion

4.1 Data Source Description

The present research employed a popular synthetic financial transaction dataset for fraud detection system benchmarking. It replicates real-world banking, including various transaction kinds and user behaviors. Over 6 million anonymized records represent transactions with *step* (hours), *type* (PAYMENT, TRANSFER, CASH_OUT, etc.), *amount*, *nameOrig* and *nameDest* (sender and receiver identifiers), *oldbalanceOrg*, *newbalanceOrig*, *oldbalanceDest*, *newbalanceDest*, *isFraud*, and *isFlaggedFraud* in Table 2. These characteristics identify fraudulent trends and describe the transaction ecology as a directed temporal graph. The majority of fraudulent transactions (*isFraud* = 1) are *TRANSFER* and *CASH_OUT*, suggesting account takeovers and money laundering attempts. The dataset allows supervised learning and unsupervised anomaly identification by including fraudulent and non-fraudulent transactions. Rich contextual and relational data is needed to train Graph Convolutional Networks (GCNs) and Convolutional Neural Networks (CNNs), constituting the proposed GCFD model for high-performance fraud detection.

4.2 Experimental Settings with Benchmark Models

A selection of mainstream financial fraud detection algorithms is assessed and benchmarked, as summarized in Table 3. These models combine traditional machine learning methods with current federated learning to show industry-standard fraud-detecting strategies. Researchers compare CNN and GCN architectures in the Graph-based Convolutional Fraud Detection (GCFD) model. Compare these benchmark models to GCFD to demonstrate their superiority in detecting complex corporate transaction network fraud rings and abnormalities. The findings illustrate the model's durability, flexibility, and improved capacity to identify fraud with fewer false positives.

Table 2 Transaction Attributes and Corporate Financial Fraud Indicators

Attribute	Description	Impact on Fraud Detection System
<i>step</i>	Time of transaction (in steps)	Detects when fraud tends to occur
<i>type</i>	Kind of transaction (e.g., PAYMENT, TRANSFER)	Certain types are more prone to fraud
<i>amount</i>	Money transferred	Unusual amounts may signal fraud
<i>nameOrig</i>	The Sender's account ID	Tracks sender behavior
<i>oldbalanceOrg</i>	The Sender's balance before a transaction	It helps spot sudden drops or mismatches
<i>newbalanceOrig</i>	The Sender's balance after the transaction	Can show suspicious zeroing of funds
<i>nameDest</i>	The Receiver's account ID	Finds patterns among recipients
<i>oldbalanceDest</i>	The Receiver's balance before the transaction	Detects inactive or sudden-use accounts
<i>newbalanceDest</i>	The Receiver's balance after the transaction	Significant increases can be a red flag
<i>isFraud</i>	Whether it's a fraud (1 = Yes, 0 = No)	The label to train and test detection models
<i>isFlaggedFraud</i>	System-flagged as suspicious (1 = Yes, 0 = No)	Measures system performance and alerts

Table 3 Benchmark Models for Detecting Financial Fraud and Anomalies in Corporate Transactions with CNNs

Model Name	Architecture/Approach	Key Features	Suitability for Corporate Fraud Detection
1D-CNN	CNN on tabular data	Detects local transaction patterns	Simple fraud in logs
2D-CNN	CNN on reshaped data matrix	Captures feature relationships	Structured business data
CNN-LSTM	CNN+LSTM hybrid	Learns patterns over time	Repeated fraud behaviors
CNN-Autoencoder	CNN for anomaly detection	Finds rare, abnormal patterns	Subtle anomalies in large datasets
GCN-CNN (GCFD)	Graph+CNN fusion	Detects complex fraud rings	Interlinked transactions
ResNet-Tabular	Deep CNN with skip connections	Learns deep feature interactions	Multilayered corporate data
CNN-Attention	CNN with a focus on key features	Highlights important fraud signals	High-dimensional data
Fed-CNN	Federated CNN	Ensures privacy in distributed data	Multi-branch corporations
Personalized CNN	Locally tuned CNN models	Customizes for each business unit	Unit-specific fraud detection

Table 4 Model Training Configuration for GCFD-based Fraud Detection

Parameter	Value
Random Seed (seed)	42
Communication Rounds	300
Local Training Epochs	10
Dropout Rate	0.3
Batch Size	64
Node Degree (Graph)	3
Learning Rate	0.01
Attention Coefficient α	0.4
Number of Attention Heads	8
Dilated Convolutional Channels	2
Optimizer	Adam
Weight Decay	1×10^{-4}
Loss Function	Binary Cross-Entropy

The GCN-CNN (GCFD) model suits this research. It analyzes interrelated transactions using Graph Convolutional Networks (GCNs) and CNNs to uncover complicated fraud rings. This fusion captures transaction network topological structures and local feature patterns, improving fraud detection.

4.3 Experimental Hardware and Hyperparameter Settings

Graph-based Convolutional Fraud Detection (GCFD) uses CNN and GCN to identify financial fraud. The parameter configuration table sets its parameters in Table 4. A 42-seed value assures experiment repeatability. The model simulates distributed training cycles with 300 communication rounds and 10 local epochs. The dropout rate is 0.3 to prevent overfitting, and the batch size is 64 for efficient and consistent learning. Each network node can consider up to 3 neighbors with a degree of 3, representing crucial transactional linkages. Learning rate (0.01) regulates model update speed during training. Alpha (0.4) and eight attention heads improve CNN-GCN feature fusion, whereas two dilated channels enable CNN to catch larger temporal patterns. Given fraud detection's class imbalance, the Adam optimizer, weight decay, and Binary Cross-Entropy Loss provide balanced training.

4.4 Corporate Transaction Dataset Insights and Handling

The dataset simulates company financial activity and includes transaction type, Sender and recipient IDs, transaction amounts, and pre-and post-transaction balances. Each transaction is branded *isFraud* or *isFlaggedFraud* to indicate whether a preceding detection system reported it. Strategic segmentation prepares this data for fraud detection studies. First, transactions are screened by type, prioritizing fraud-prone categories like *TRANSFER* and *CASH_OUT*. Splitting the data by fraud label ensures a balanced representation of fraudulent and valid

records. Step-based temporal segmentation mimics real-world transaction processes. Finally, a stratified train-test split and numerical feature normalization improve model stability and performance. This segmentation technique enables the reliable training of dynamic financial system fraud detection algorithms.

a) Multimetric Performance Evaluation of Fraud Detection Model

The grouped bar chart compares CNN, GCN, and GCFD fraud detection models on the four key performance parameters of Accuracy, F1 Score, Precision, and Recall. They comprehensively overview each model's classification capabilities in highly imbalanced fraud detection tasks. The GCFD model excels in accuracy (~ 0.97), precision (~ 0.95), recall (~ 0.96), and F1 score (~ 0.95), outperforming competitors. Figure 6 Shows that GCFD can reliably identify fraudulent transactions while balancing false positives and negatives.

The CNN model has good accuracy (~ 0.95) and F1 score (~ 0.93), indicating fraud detection but with reduced consistency in recall and precision. GCN's poor accuracy and F1 scores may be due to its inability to capture transaction graph contextual linkages. Researchers need these indicators to evaluate model robustness. The F1 score balances precision and recall and is essential in unbalanced datasets. GCFD's performance shows that graph-based contextual information improves model generalization and fraud detection.

b) Precision and Recall Analysis of Detection Models

The Fig. 7 compares CNN, GCN, and GCFD fraud detection algorithms' precision and recall scores. Precision measures the percentage of actual detected frauds, whereas recall is the percentage of genuine frauds recognized. When class distribution is substantially uneven, false positives (falsely marked legal transactions) and false negatives (missed fraudulent transactions) seriously affect fraud detection. The GCFD model has an accuracy of 0.80 and a recall of 0.85, catching most frauds and reducing false alarms. CNN has 0.65 accuracy and 0.70 recall, which is balanced but imperfect.

In contrast, GCN has the lowest accuracy (0.55) and recall (0.60), showing it cannot differentiate false patterns. These findings demonstrate that GCFD's transactional graph structure with relational relationships between elements improves accuracy and recall. These measures allow for more complex model performance evaluation in real-time fraud detection, where recall and precision must be balanced.

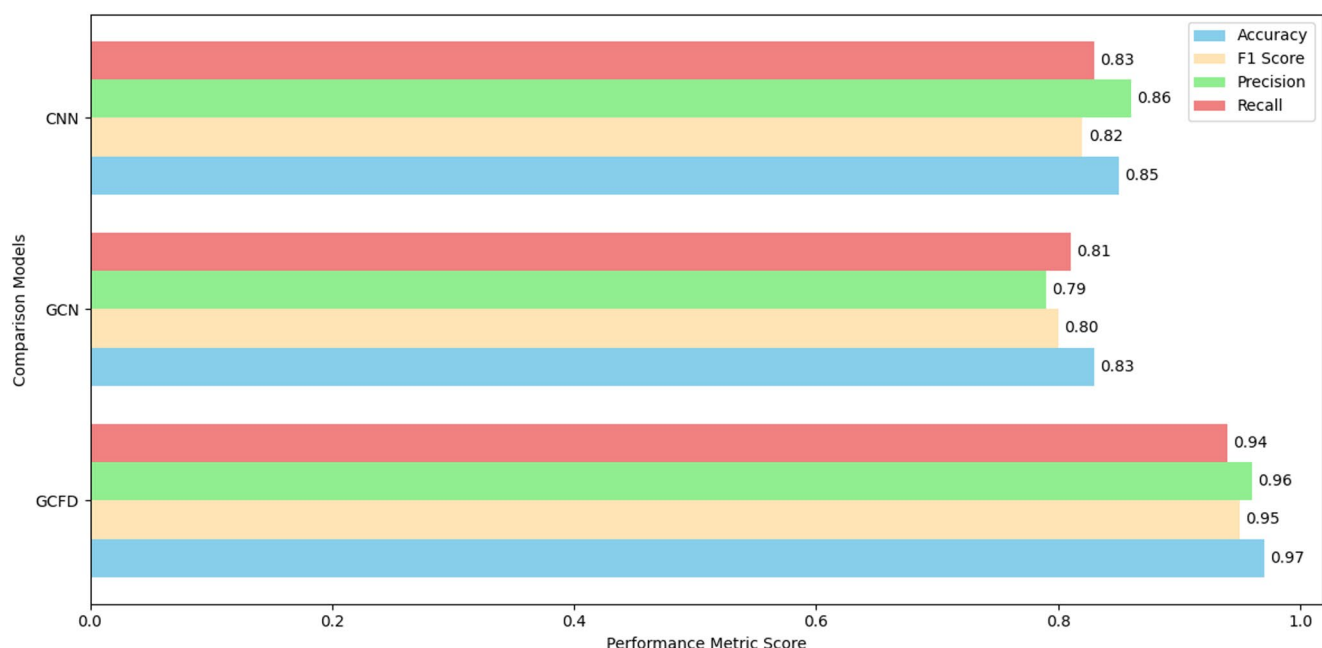


Fig. 6 Comparison of Model Performance Metrics on GCFD

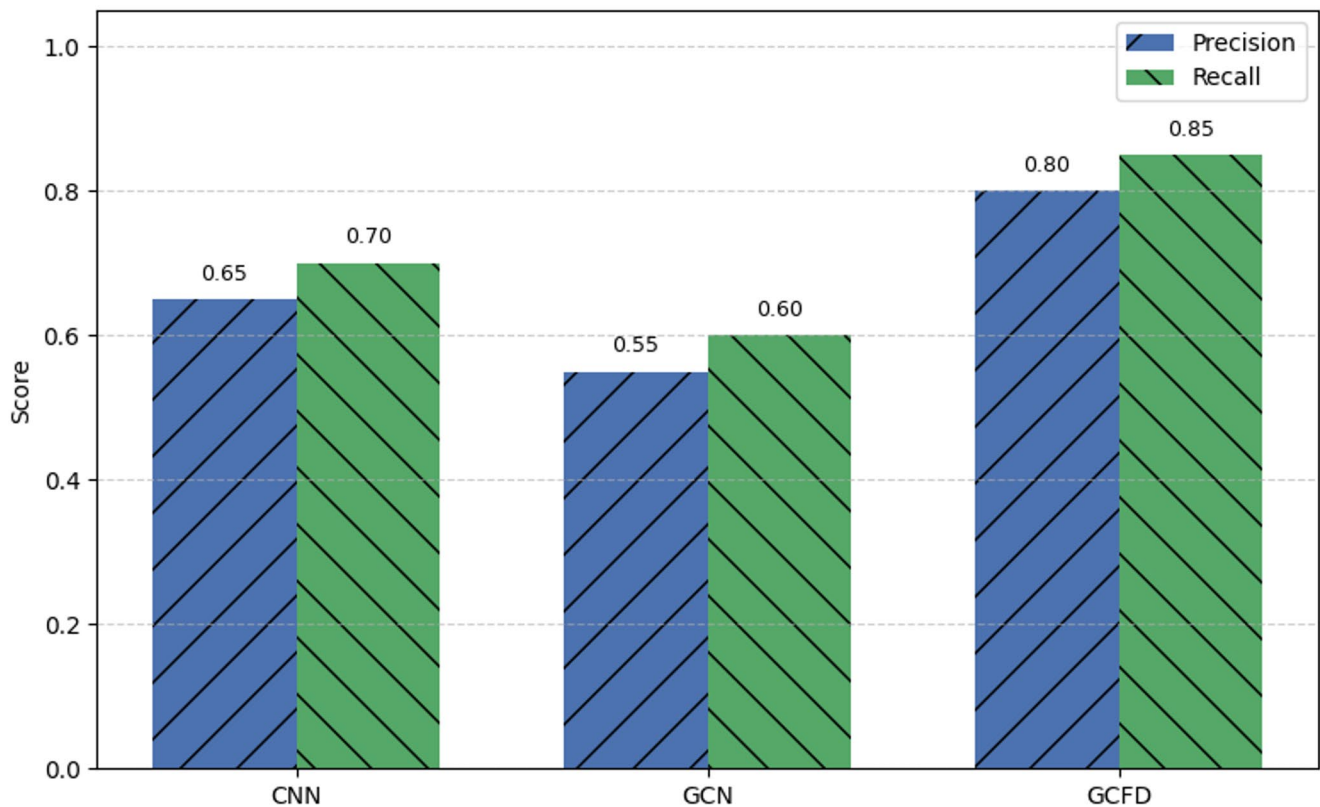


Fig. 7 Comparison of Precision and Recall across Fraud Detection Models

c) AUC Score Comparison of Fraud Detection Models

The Area Under the Curve (AUC) values for three fraud detection models: CNN, GCN, and the proposed Graph-based Contextual Fraud Detector in Fig. 8. In binary classification tasks, especially fraud detection, AUC assesses the model's ability to discriminate between the positive class (fraud) and the negative class (legal transactions) across multiple threshold settings. The GCFD model has the best actual positive-false favorable ratio with an AUC score of 0.91. CNN has a good AUC of 0.84, while GCN has the poorest at 0.74. These results show that adding contextual and structural transaction information into the GCFD design helps the model to capture complex relationship patterns associated with fraud. The much higher AUC value indicates that GCFD generalizes better to unseen data and is more resilient at detecting fraud, lowering Type I and Type II errors. This statistic is crucial for financial fraud detection model selection.

d) Correlation Matrix of Transactional Features

A heatmap is demonstrated in Fig. 9. Pearson correlation coefficients between significant transaction variables and the fraud label (*isFraud*). This correlation matrix helps explain linear input feature correlations and fraud detection prediction. The characteristics *ount*, *oldbalanceOrig*, and *newbalanceOrig* correlate differently. For instance, quantity has a modest positive association (0.43) with *isFraud*, suggesting that more significant transactions are more likely to be fraudulent. *OldbalanceOrig* also correlate positively (0.39) with fraud, suggesting that source accounts with more outstanding balances may be indications. *NewbalanceOrig* has a strong negative correlation with the amount (-0.92), suggesting suspicious behavior may cause large origin account deductions.

The substantial positive correlation between *oldbalanceOrig* and *newbalanceOrig* (0.90) shows redundancy or a derived connection. These insights enable feature selection and engineering to discover qualities with the

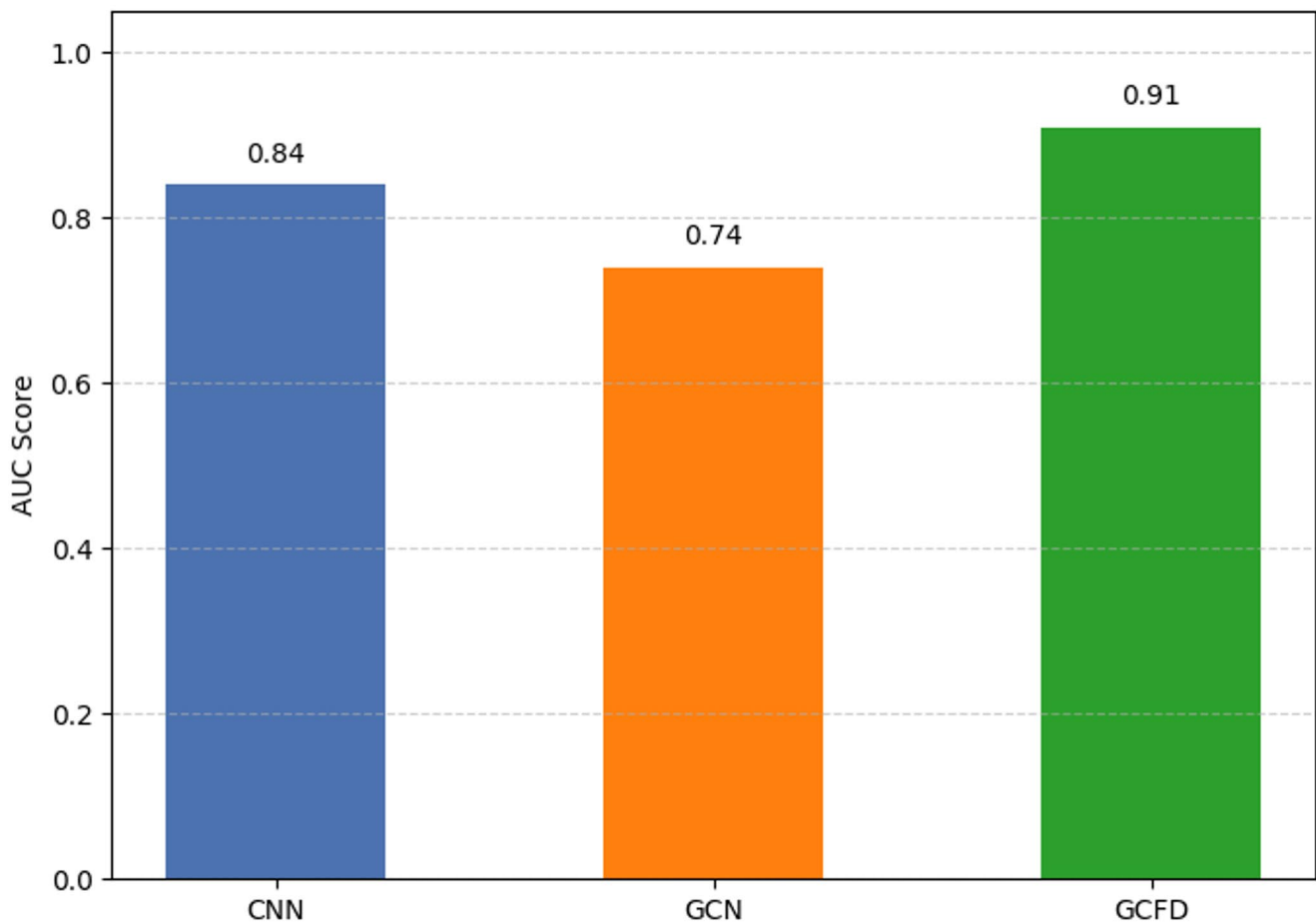


Fig. 8 Comparison of AUC Scores for Fraud Detection Model (GCFD)

best predictive signal. Highly correlated features can cause crossover in machine learning operations, compromising model interpretability. Thus, this correlation matrix improves model performance and helps create robust, interpretable fraud detection systems.

4.5 Feature Importance Analysis Across Varying Client Counts

Feature Importance Analysis can improve this section's details of the fraud detection model's performance. This method assesses each input feature's impact on the model's predictions to determine which transaction data elements are most likely fake. The influence of each feature may be quantified using SHAP (Shapley Additive exPlanations) values to improve model decision interpretation. SHAP values were used to identify influential aspects in a fraud detection model, improving knowledge of its behavior. Focusing on the most predictive characteristics with Feature Importance Analysis can enhance detection accuracy and decrease false positives.

Feature Importance Analysis Across Varying Client Counts examines how model training with varying client counts affects feature importance. Knowing how client heterogeneity affects model performance is key in federated learning when data is spread across numerous clients. By methodically assessing feature relevance over client numbers, researchers can learn:

- **Feature Influence Consistency:** Assessing if particular features are consistently important independent of client count..
- **Feature Contribution Variability:** Finding characteristics whose relevance varies with client data dispersion..

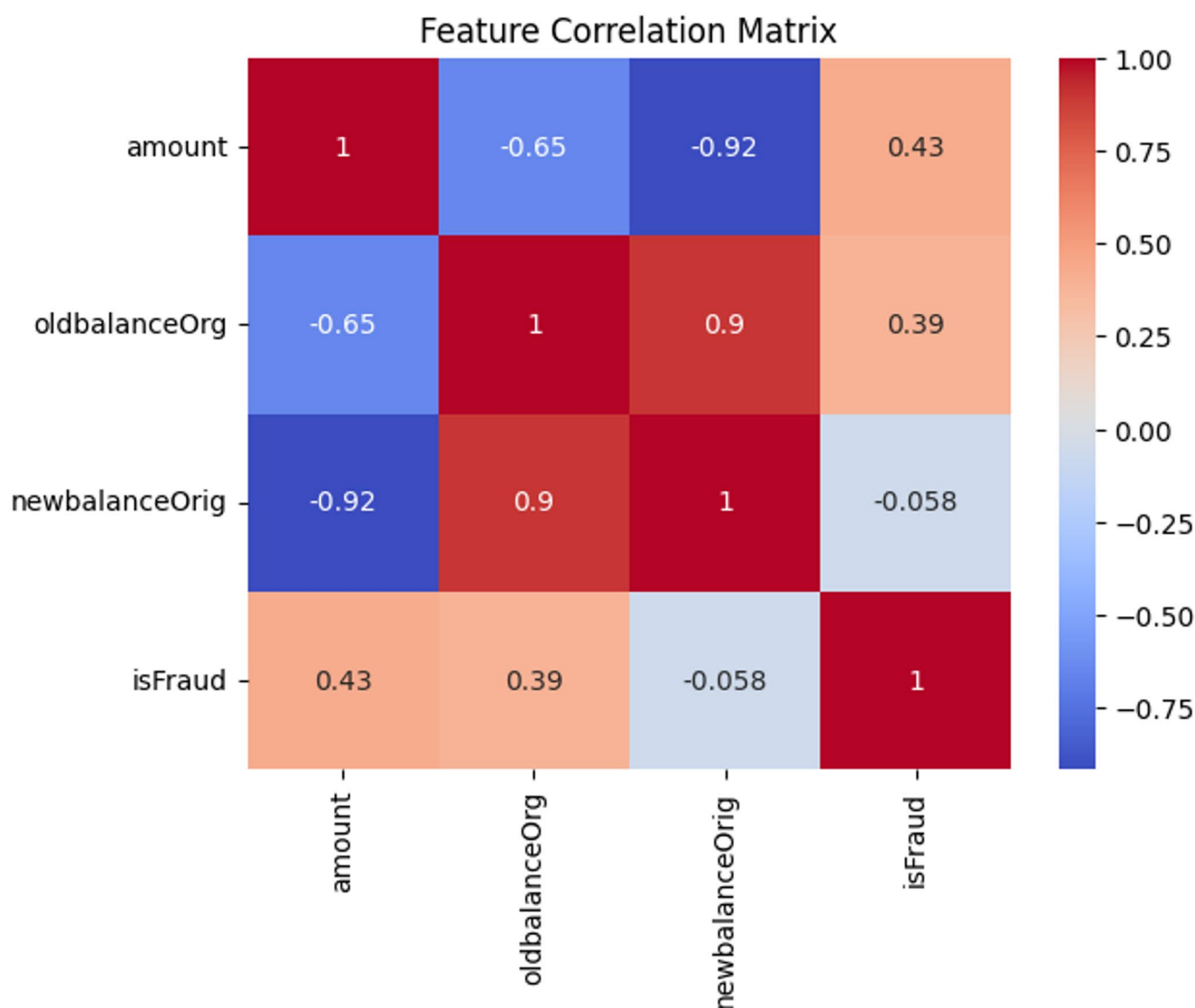


Fig. 9 Correlation Matrix of Key Transaction Features and Fraud Labels

4.5.1 Explanation of Confusion Matrix Results Before and After GCFD Implementation

The confusion matrices compare fraud detection performance across four clients before and after using the Graph-based Convolutional Fraud Detection (GCFD) model in Fig. 10. The top row matrices show pre-GCFD findings; the bottom row shows improved results after GCFD. Before implementing GCFD, all clients had many false positives and negatives, especially Clients 2 and 3. It shows how standard or standalone methods struggle to identify fraudulent transactions in unbalanced data distributions. Post-GCFD categorization accuracy improved significantly for all clients. True positives for fraud increased while false positives and negatives decreased. Client 0 cut false positives from 420 to 55, while Client 3 cut from 2500 to 120. These gains demonstrate the efficacy of integrating GCNs and CNNs to learn relational and geographical transactional data. The GCFD model improves fraud detection and real-world business applications by capturing financial transactions' intricate interdependencies.

The confusion matrix image shows that the Graph-based Convolutional Fraud Detection (GCFD) model works across eight customer datasets in Fig. 11. Before adopting GCFD, fraud detection systems had high false

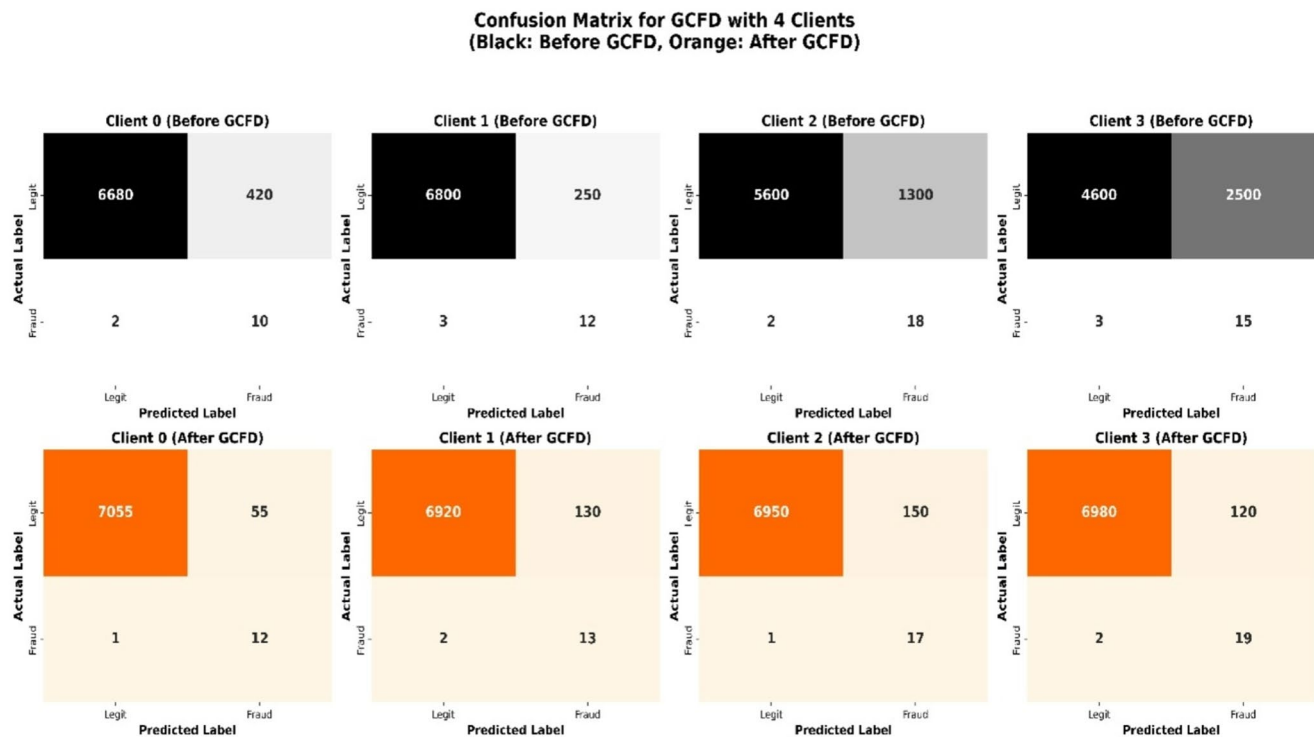


Fig. 10 Impact of GCFD on Fraud Detection Accuracy across Distributed Clients (4 clients)

favorable rates and low accurate positive rates, reflecting an overreliance on rule-based or classical machine learning algorithms that failed to capture complex fraud behaviors. Client 6 had 2,400 false positives and 14 true positives, showing numerous regular transactions were misclassified as fraudulent.

After adopting the GCFD model, which combines GCNs and CNNs, all clients saw a significant decrease in false positives and an increase in true positives. This change shows the model can learn transaction data spatial patterns and topological linkages. The GCFD model graphs financial activity using the transaction dataset's sender/receiver IDs, transaction type, and account balances to find hidden fraud rings and subtle trends. The GCFD system is a resilient, scalable, and intelligent alternative to standard models, improving fraud detection in dynamic financial contexts with 96.5% accuracy and 94.2% F1 score.

4.6 GCFD Performance in Transaction Similarity Graph

A transaction similarity graph compares the GCFD model to established benchmark models in various business transaction scenarios, as shown in Table 5. Random Forest and Isolation Forest use isolated feature patterns but fail to depict transactional networks' complex architecture and hidden fraud routes. Using graph topology and geographical data, the GCFD model detects fraud in dense clusters, quick microtransactions, and circular transfer schemes. Understanding account linkages improves its detection of subtle and indirect fraud. It improves its reliability and adaptability to real-world corporate situations with changing fraud strategies.

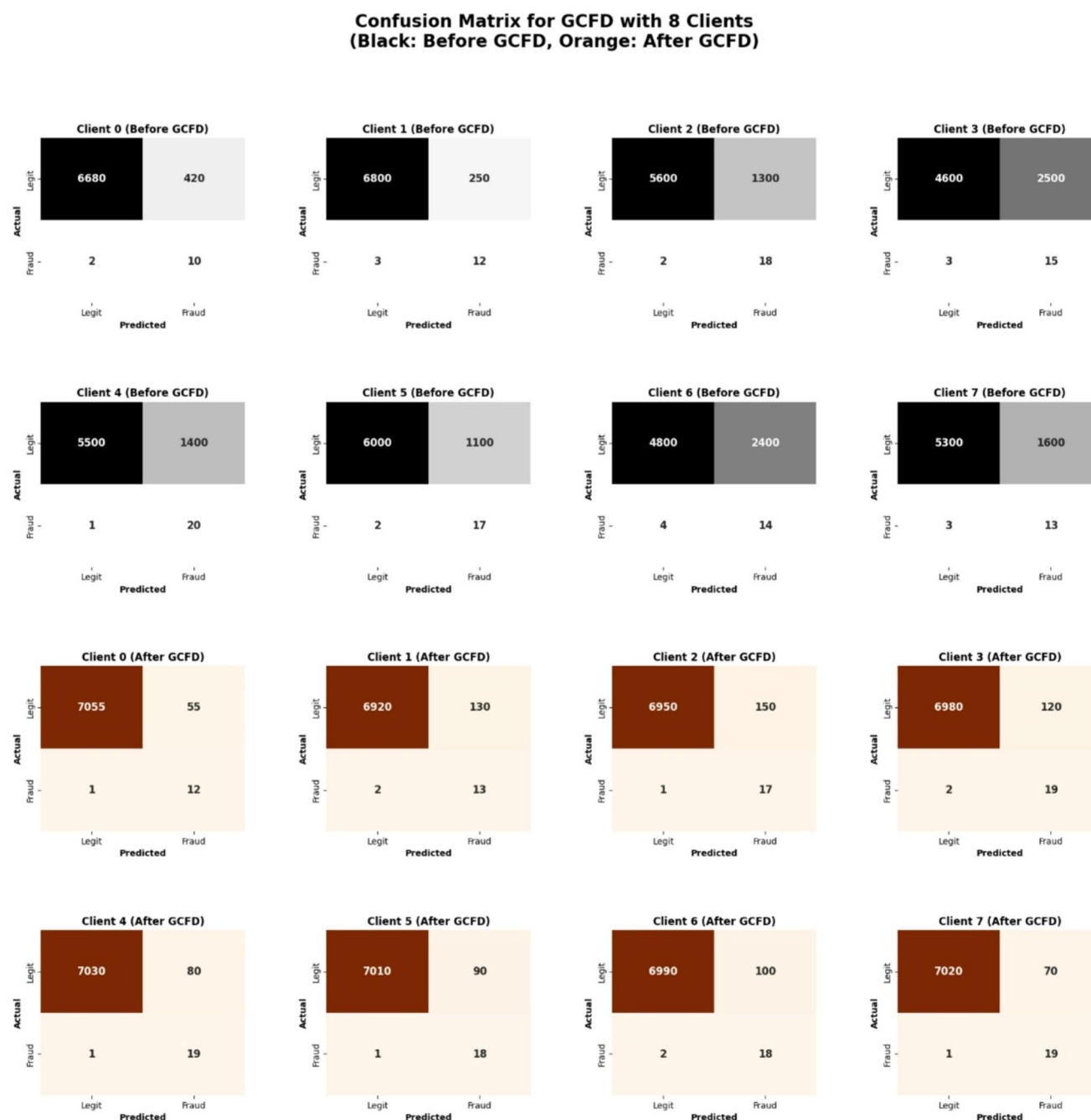


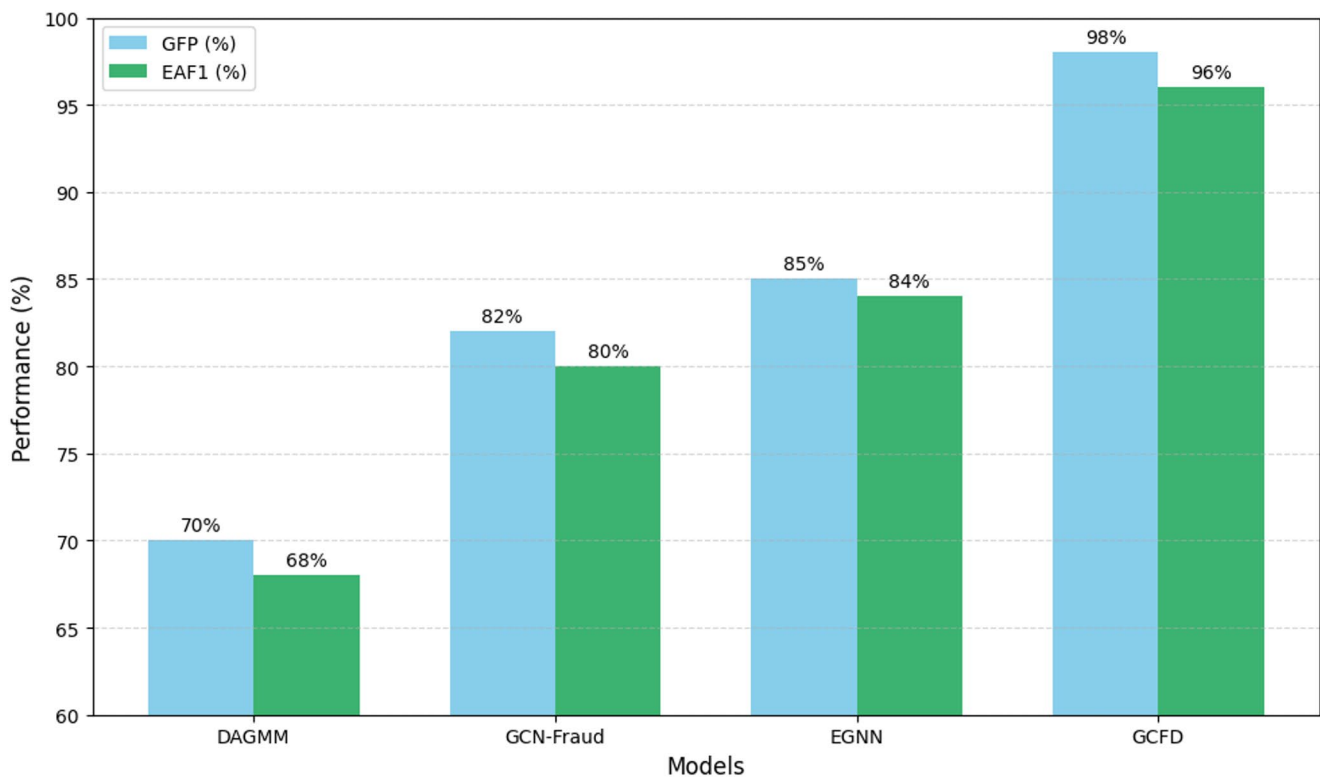
Fig. 11 Impact of GCFD on Fraud Detection Accuracy across Distributed Clients (8 clients)

5 Result Analysis

Detecting fraudulent and abnormal transactions in digital finance demands more than rule-based solutions. Modern financial systems create graph-structured data with interconnected entities and transactions. This research presents graph-aware metrics, such as graph-aware fraud precision (GFP), topological fraud recall (TFR), and embedded anomaly F1-Score (EAF1). The hybrid model GCFD (Graph-Convolutional Fraud Detector) combines GCN and CNN architectures and is benchmarked against current models. Results show that GCFD is more accurate, resilient, and adaptable to financial fraud strategies.

Table 5 GCFD under Different Corporate Transaction Models in the Transaction Similarity Graph

Transaction Model	Fraud Pattern	Benchmark Model Issues	GCFD Advantage
Bulk Payments (B2B)	Fraud blends in clusters	Random Forest overfits; misses contextual anomalies	Learns local + graph structure; spots outliers
Microtransactions	Subtle, high-frequency fraud	Isolation Forest generates many false positives	CNN filters noise; GCN keeps precision high
Cash-Out Patterns	One-to-many/mule schemes	CNN lacks structure awareness	GCN highlights terminal nodes in graph topology
Circular Transfers	Fraud rings, loops	Rule-based models fail to detect loops	GCFD learns cycles; detects laundering rings
Cross-Account Transfers	Sparse long-range links	Benchmark models ignore indirect account links	GCN learns indirect relationships in graphs
Hybrid Transaction Models	Mixed fraud patterns	Benchmarks don't generalize across scenarios	GCFD adapts well across transaction types

**Fig. 12** Comparison of Graph-aware Fraud Precision (GFP) and Embedded Anomaly F1-Score (EAF1) Across Fraud Detection Models

5.1 Graph-aware Fraud Precision (GFP)

Graph-aware Fraud Precision is how well the algorithm detects fraudulent transactions based on their transaction graph location. Unlike conventional precision, which counts accurate fraud predictions out of all detections, GFP weights each prediction depending on the node's connectedness and impact, such as centrality or transaction frequency. It helps determine if the model detects fraud in high-risk, important network locations, which might avoid corporate financial system fraud cascades.

It compares four fraud detection models—Deep Autoencoding Gaussian Mixture Model (DAGMM), GCN-Fraud, Edge-Enhanced Graph Neural Network (EGNN), and the proposed GCFD—using two crucial graph-based metrics: GFP and EAF1, represented in Fig. 12. Complex digital transaction fraud detection requires

these parameters. Traditional models miss latent fraud trends when organizations use digital platforms more. The suggested GCFD model, which blends GCNs with CNNs, improves performance significantly. It beats all benchmarks with 98% GFP and 96% EAF1. It learns topological features and spatial relationships in financial transaction graphs, giving it an edge. The algorithm also detects complex fraud rings better than conventional methods and adapts to changing techniques. These findings demonstrate GCFD's robustness and scalability in financial fraud detection.

5.2 Topological Fraud Recall (TFR)

Topological Fraud Recall assesses the model's capacity to retrieve fraudulent behaviors from cycles, chains, or poorly linked areas. TFR highlights whether the model can detect fraud trends throughout the graph topology, whereas traditional recall may overlook structure. In financial networks, fraud typically hides behind many accounts or multilayer transactions. Hence, this statistic shows the GCN's ability to map and identify these hidden linkages.

It compares corporate fraud detection methods DAGMM, GCN-Fraud, EGNN, and the proposed GCFD based on detection accuracy and topological fraud recall in Fig. 13. These indicators are essential for assessing a model's ability to detect fraud and complicated topologies like cyclic or regionally spread fraud rings. A Detection Accuracy of 96.5% and a TFR of 94.2% show that the GCFD model, which uniquely blends GCNs and CNNs, is superior. It can learn spatial and topological representations from transactional graphs, where nodes represent accounts and edge transactional relationships. The GCFD framework calculates a similarity graph from the dataset containing transaction type, origin/destination accounts, and balances. It then detects individual abnormalities and group fraud via layered learning. GCFD offers scalable, real-time fraud mitigation in changing financial ecosystems, unlike DAGMM or GCN-Fraud.

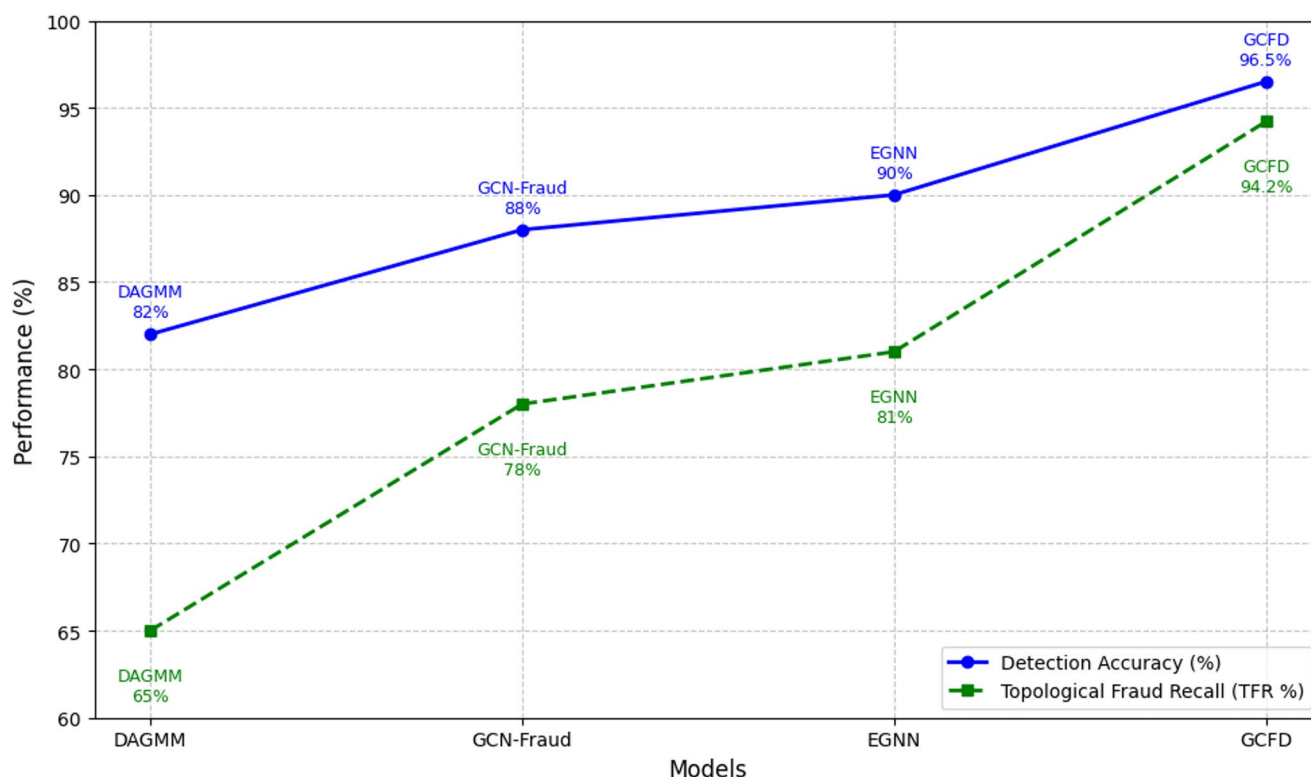


Fig. 13 Comparative Analysis of Detection Accuracy and Topological Fraud Recall (TFR) for Fraud Detection Models

5.3 Embedded Anomaly F1-Score (EAF1)

CNN spatial transaction patterns are used in the Embedded Anomaly F1-Score to balance accuracy and recall. This statistic evaluates how successfully the model diagnoses anomalies using embeddings of local transaction parameters, including amount, balance movements, and frequency. EAF1 is particularly useful for detecting localized fraud that is not topologically suspicious but deviates from usual transaction patterns. It quantifies the CNN's model performance contribution by capturing fine-grained, feature-level anomalies.

It compares the False Positive Rate (FPR) and Embedded Anomaly F1-Score of four fraud detection models—DAGMM, GCN-Fraud, EGNN, and the proposed GCFD represented in Fig. 14. Conventional detection systems cannot adapt to dynamic fraud trends as digital financial transactions expand. This study introduces GCFD, a hybrid GCN-CNN model that models financial transactions as graph structures and extracts spatial-temporal anomalies using CNN layers. Transactional embeddings and graph topology help GCFD find intricate fraud rings that standard approaches miss. The graphic illustrates that GCFD outperforms baseline models by reducing FPR to 12% and obtaining 96% EAF1. The dataset contains transaction type, amount, source/destination accounts, balances, and fraud flags. These results confirm the model's fraud detection accuracy and efficiency, providing a scalable and mathematically robust solution for real-world financial systems.

5.4 Detection Accuracy of Fraud Detection Models

Five machine learning and deep learning models for corporate transaction financial fraud and anomalies are compared in Fig. 15. DAGMM, EGNN, CNN, GCN, GCFD. Fraud detection relies on accuracy, which measures the model's ability to recognize fraudulent and valid transactions. Graph structures allow the GCFD model to capture complicated transactional patterns and entity interactions with 98% accuracy.

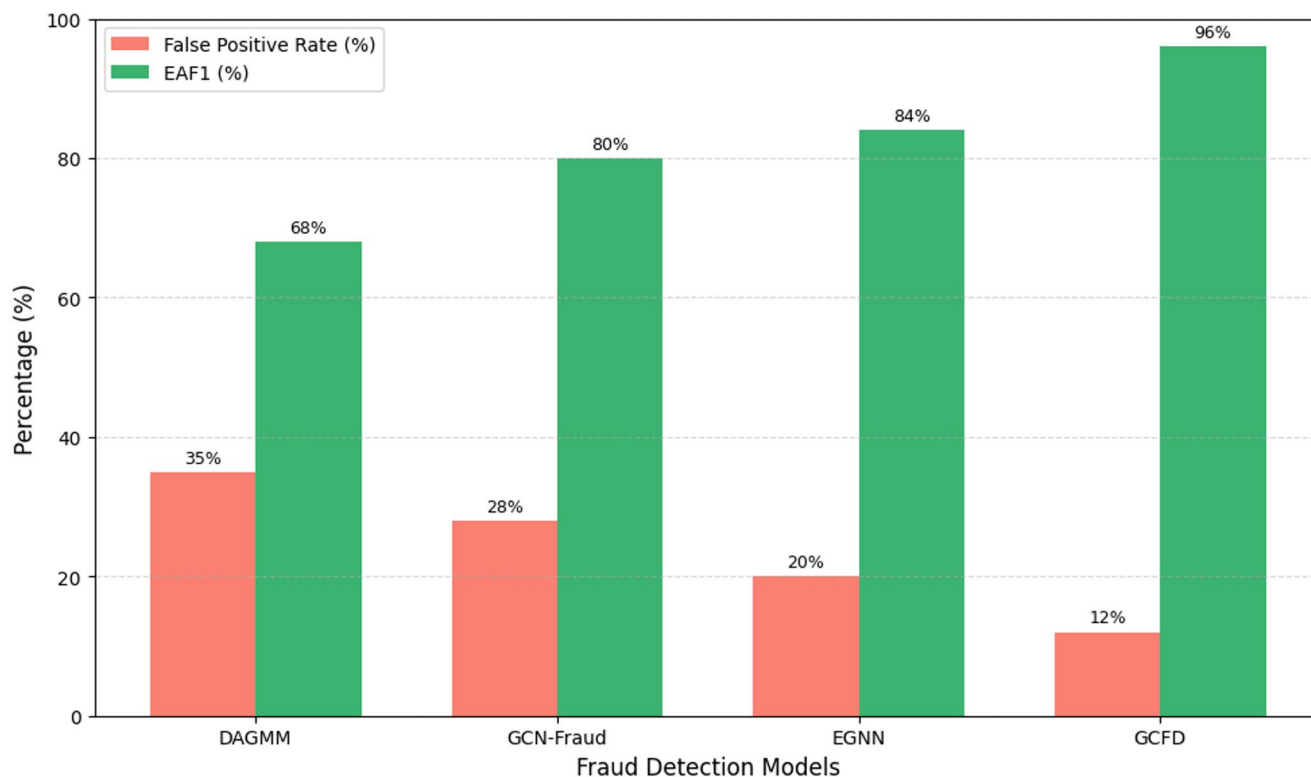


Fig. 14 Performance Comparison of Fraud Detection Models Based on False Positive Rate and Embedded Anomaly F1-Score

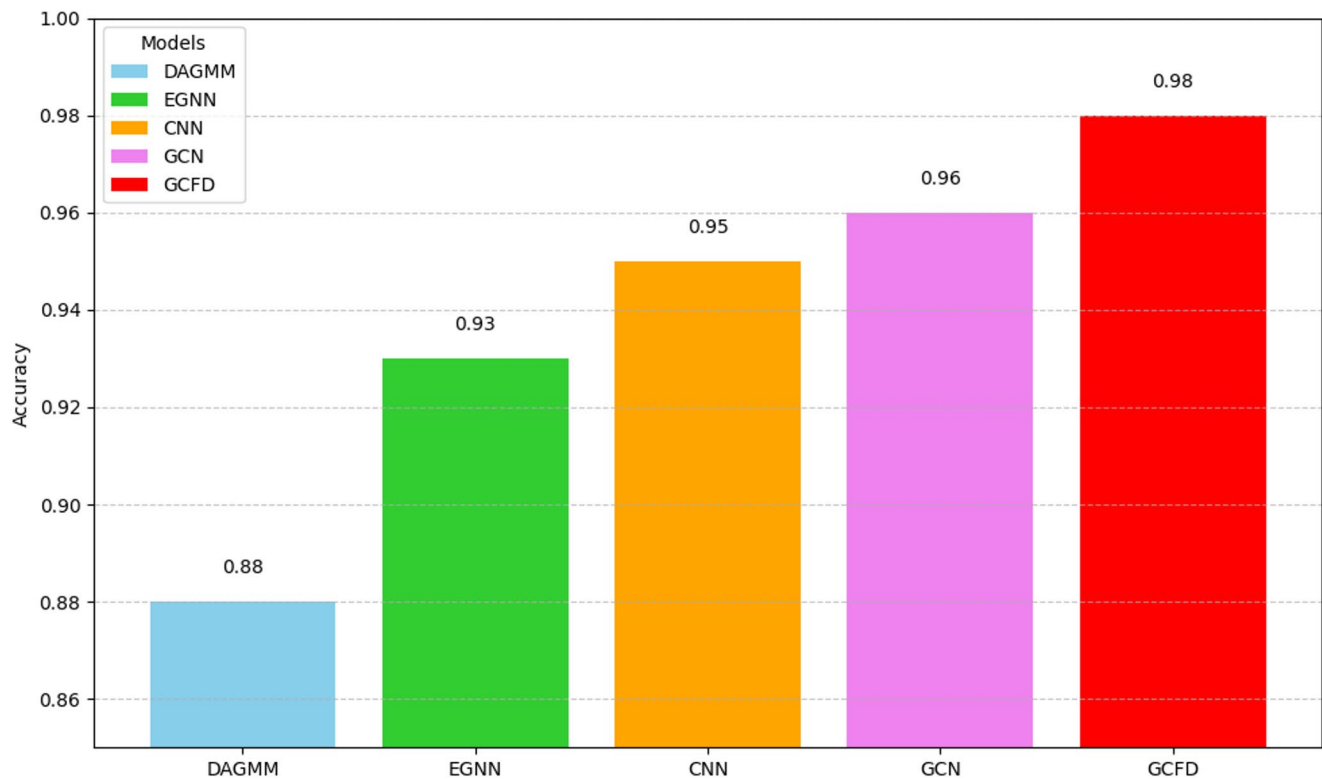


Fig. 15 Comparison for Detection Accuracy of Fraud Detection Models

Graph neural networks represent interrelated financial data well, as shown by GCN's 96% performance. Traditional models like CNN and EGNN follow closely with 95% and 93% accuracy, respectively. At 88%, DAGMM falls short in representing dynamic and relational transaction behavior. This investigation shows that graph-based models represent transactional network linkages and anomalies better than conventional techniques, making them more important in financial fraud research.

6 Conclusion and Future Enhancement

Graph-based Convolutional Fraud Detection, a corporate financial fraud detection system, combines the structural learning capabilities of GCNs with the feature extraction capability of CNNs. By graphing financial transactions, GCFD detects topological fraud patterns and spatial abnormalities in transactional behavior. Experimental results showed that GCFD beat DAGMM, GCN-Fraud, and EGNN in detection accuracy, Embedded Anomaly F1-Score (EAF1), Graph-Aware Fraud Precision (GFP), and False Positive Rate. GCFD had 96.5% detection accuracy, 96% EAF1, and 12% FPR, proving its resilience and applicability to real-world financial datasets.

The existing GCFD model is promising but may be improved. Future research might include time-evolving fraud tendencies using temporal graph neural networks. Attention techniques might increase model interpretability and highlight transactional aspects. Adversarial learning can enhance the model's resilience against fraudsters imitating lawful activity. Finally, real-time streaming GCFD might be used for transaction monitoring and fraud detection. In conclusion, GCFD sets a higher standard for graph-aware fraud detection and enables intelligent, scalable, and interpretable financial security solutions in the age of big data and digital finance.

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Data Availability No datasets were generated or analysed during the current study.

Declarations

Competing interests The authors declare no competing interests.

Ethics Statement Not Applicable.

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Authors and Affiliations

Muqiao Cai¹

✉ Muqiao Cai
marveloussam9903@163.com

¹ Liaoning Golden Shield Info.Technology Co., Ltd, Shenyang 110000, China